AEGIS-AI

ITC-2026-C0-1637

Adaptive Electronic Guard & Intelligence System

Anti-Drone Competition 2026

Air Defense College | 6th ITC International Conference

April 2026

Multi-Agent Swarm Intelligence for Counter-UAS Tracking & Behavior Detection (Swarm vs Swarm)

- Tier-1: Extended Kalman Filter multi-target tracking
- Tier-2: Reynolds inversion + GNN behavior classification
- Tier-3: PPO response coordinator
- 3-source sensor fusion: RF · Acoustic · Vision/EO-IR

500+ Drones tracked	**3** Sensor types	**6** Behavior classes	**<100ms** End-to-end

The Challenge: Drone Swarms vs. Traditional Defense

150 : 1

Cost asymmetry — attacker vs. defender

$20K suicide drone vs $3M interceptor

Asymmetric Cost

Each $20K drone forces a $3M response.
Swarms deliberately drain missile inventories.

Emergent Behavior

Swarm tactics emerge from local rules.
Coordinated maneuvers: pincer, saturation, decoy.

Adaptive Learning

Destroyed drones transmit defense positions.
Surving drones route around defensive fire.

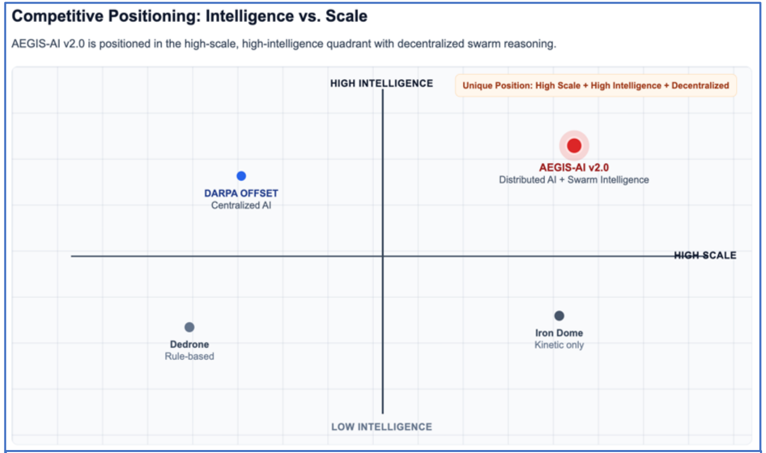
The Gap: No existing C-UAS system models the swarm as a collective entity with emergent intelligence. Individual-tracker approaches are fundamentally inadequate.

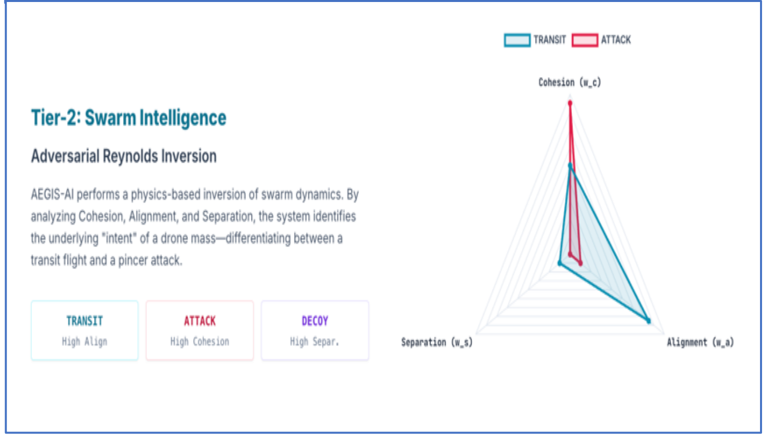
Current C-UAS Generations

Gen-1 Kinetic:	Ballistic — one target at a time
Gen-2 EW/RF:	Signal jamming — defeated by INS nav
Gen-3 AI (single):	**Single-target AI — no swarm model**

Competitive Landscape Decomposition

Category	Representative Systems	Core Approach	Key Shortcoming	AEGIS-AI Differentiation
A. Single-Target Kinetic Defense	Iron Dome, Patriot PAC-3, SkyHunter	One-interceptor-per-drone	150:1 cost asymmetry; inventory exhaustion against swarms	Tier-3 PPO: Economic optimization under cost constraints
B. RF/EW Jamming Systems	DroneShield, AUDS, Kaspersky Antidrone	Signal jamming / GPS spoofing	Defeated by INS/GPS-denied navigation; collateral interference	Tier-1: Multi-sensor fusion (RF + Acoustic + Vision) with CI
C. Single-Target AI Trackers	YOLO + KF pipelines, OpenCV trackers	Deep learning detection + classical filtering	No swarm model; track swaps during crossing; no intent inference	Tier-1: Hungarian LAPJV + Tier-2: DBSCAN + Reynolds + GNN
D. Centralized Swarm Detection	**DARPA OFFSET, ALIAS**	**Centralized planning for swarm coordination**	**Single point of failure; latency at scale; no real-time behavior classification**	**Tier-2: Distributed graph inference with attention-based leadership identification**
E. Academic Swarm Models	Reynolds simulators, Vicsek models	Physics-based flocking simulation	No adversarial inversion; no tactical intent classification	Tier-2: MLE inversion of Reynolds weights → 6-class behavior taxonomy
F. Commercial C-UAS Platforms	Dedrone, Fortem SkyDome, Rafael Dome	Multi-sensor integration with rule-based response	Static threat rules; no learning-based CM allocation; no cost-awareness	Tier-3: PPO with curriculum learning for 150:1 asymmetry





AEGIS-AI: Fight *Swarm Intelligence* with *Swarm Intelligence*

📡 TIER 1 — Individual Tracking

- Extended Kalman Filter (CTRA model)
- Hungarian data association (LAPJV)
- 0.05ms per drone · 500 drones @ 10Hz

🧠 TIER 2 — Swarm Intelligence

- DBSCAN swarm grouper (vel-weighted)
- Reynolds parameter inversion (MLE)
- GNN behavior classifier · 6 classes

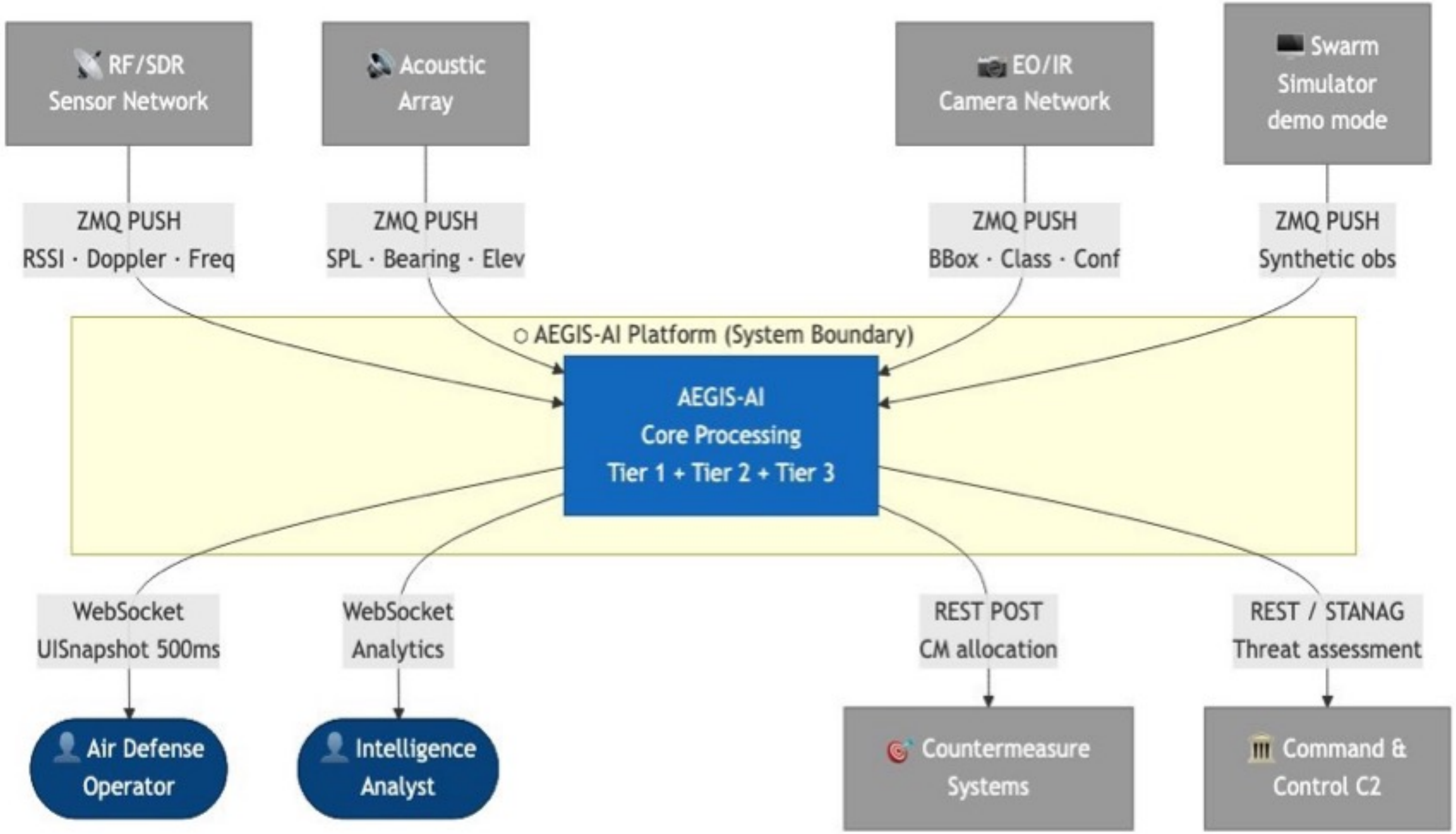
⬡ AEGIS-AI

🎯 TIER 3 — Response Coordinator

- PPO reinforcement learning policy
- Composite threat scoring (4-factor)
- CM allocation under constraints

Core Thesis: The decisive advantage is not weapons power — it's the speed and accuracy of the collective decision algorithm.

AEGIS-AI: System Context



Architecture View

EXTERNAL

- RF/SDR
- Acoustic
- Vision EO/IR
- Simulator
- Operators
- CM Systems

AEGIS-AI Platform

Ingest Service
Python/ZMQ

Tracker Pool ×2
Python/NumPy EKF

Swarm Intel
GNN + Reynolds

Coordinator
PPO/SB3

Redis Streams
In-memory bus · 5 streams

PostgreSQL
Track history

API Gateway
FastAPI/WebSocket

Tactical UI
5 pages: Radar · Swarm Graph · Behavior · Scenarios · Analytics

→ *ZMQ PUSH (all sensors)*
→ *Redis XADD/XREAD · Consumer groups · 10Hz / 2Hz / 1Hz*
→ *WebSocket ws:// → UISnapshot 500ms | REST POST → CM Systems*

Multi-Source Sensor Fusion — Covariance Intersection

RF / SDR

Range:	**4,000m**
Pos. σ:	**±15m**
Miss prob:	**3%**

RSSI dBm · Doppler m/s · Freq MHz

69,791 obs (SCN-05)

Acoustic Array

Range:	**1,200m**
Pos. σ:	**±8m**
Miss prob:	**5%**

SPL dB · Bearing° · Elevation° · Rotor Hz

42,593 obs (SCN-05)

Vision / EO-IR

Range:	**1,500m**
Pos. σ:	**±3m**
Miss prob:	**4%**

Class · Confidence · BBox px · Thermal

53,626 obs (SCN-05)

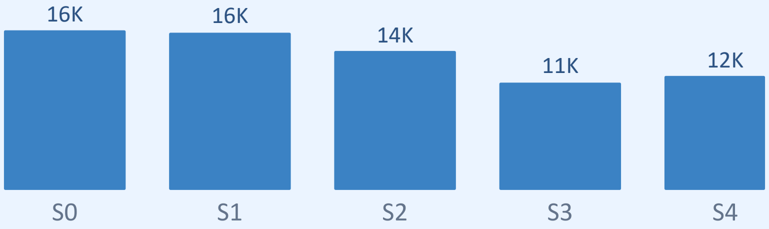
Covariance Intersection Fusion

$$P^{-1}\_fused = \omega \cdot P_1^{-1} + (1-\omega) \cdot P_2^{-1}$$

$$\hat{x}\_fused = P\_fused \cdot (\omega \cdot P_1^{-1} \cdot \hat{x}_1 + (1-\omega) \cdot P_2^{-1} \cdot \hat{x}_2)$$

ω = argmin det(P_fused) → guaranteed consistency, no overconfidence

Sensor Nodes — SCN-05 Obs per Node



Mathematical Foundation

📐 Extended Kalman Filter (CTRA)

$$x = [p_x, p_y, p_z, v_x, v_y, v_z, \omega]^T$$
$$K = P \cdot H^T \cdot (H \cdot P \cdot H^T + R)^{-1}$$
$$\hat{x} = \hat{x}^- + K \cdot (z - H \cdot \hat{x}^-)$$

~0.05ms per drone

🌊 Reynolds Flocking Inversion

```
v_i = w_s·f_sep + w_a·f_align + w_c·f_coh
MLE: argmin ||v_model - v_obs||²
→ BehaviorClass + confidence
```

6-frame history window

🔗 Graph Attention Network

$$\alpha_{ij} = \mathrm{softmax}(\mathrm{LeakyReLU}(a^T [Wh_i \| Wh_j]))$$
$$h_i^{l+1} = \sigma(\Sigma_j\, \alpha_{ij} \cdot W^l \cdot h_j^l)$$
$$z = \Sigma_i\, \mathrm{MLP}(h_i^L) \rightarrow \text{6-class}$$

Graph Attention layers × 3

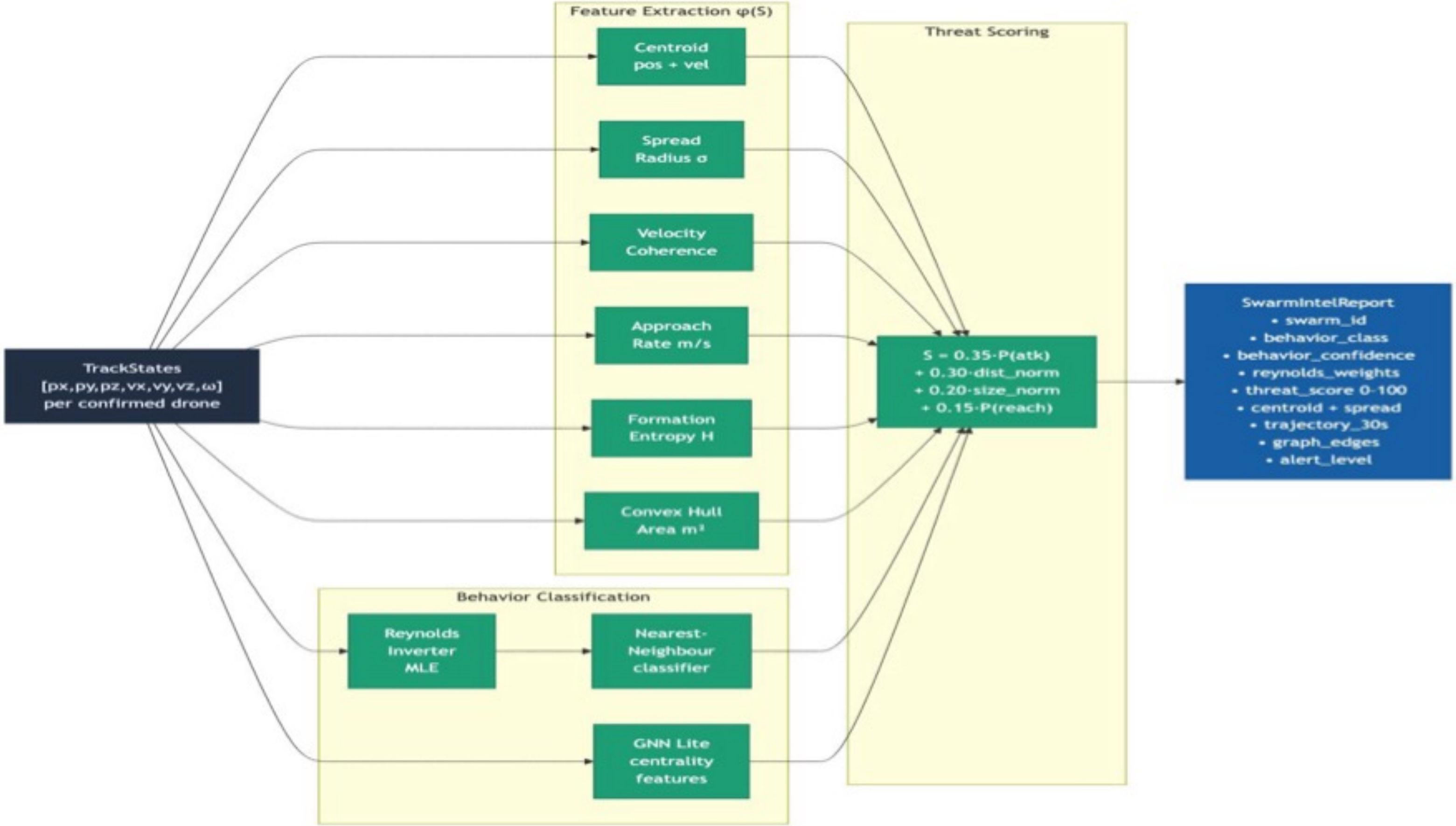
⚡ Threat Scoring & PPO

$$S = 0.35 \cdot P(atk) + 0.30 \cdot dist + 0.20 \cdot size + 0.15 \cdot P(reach)$$
$$R_t = \Sigma\, neutralized - \lambda \cdot cost$$

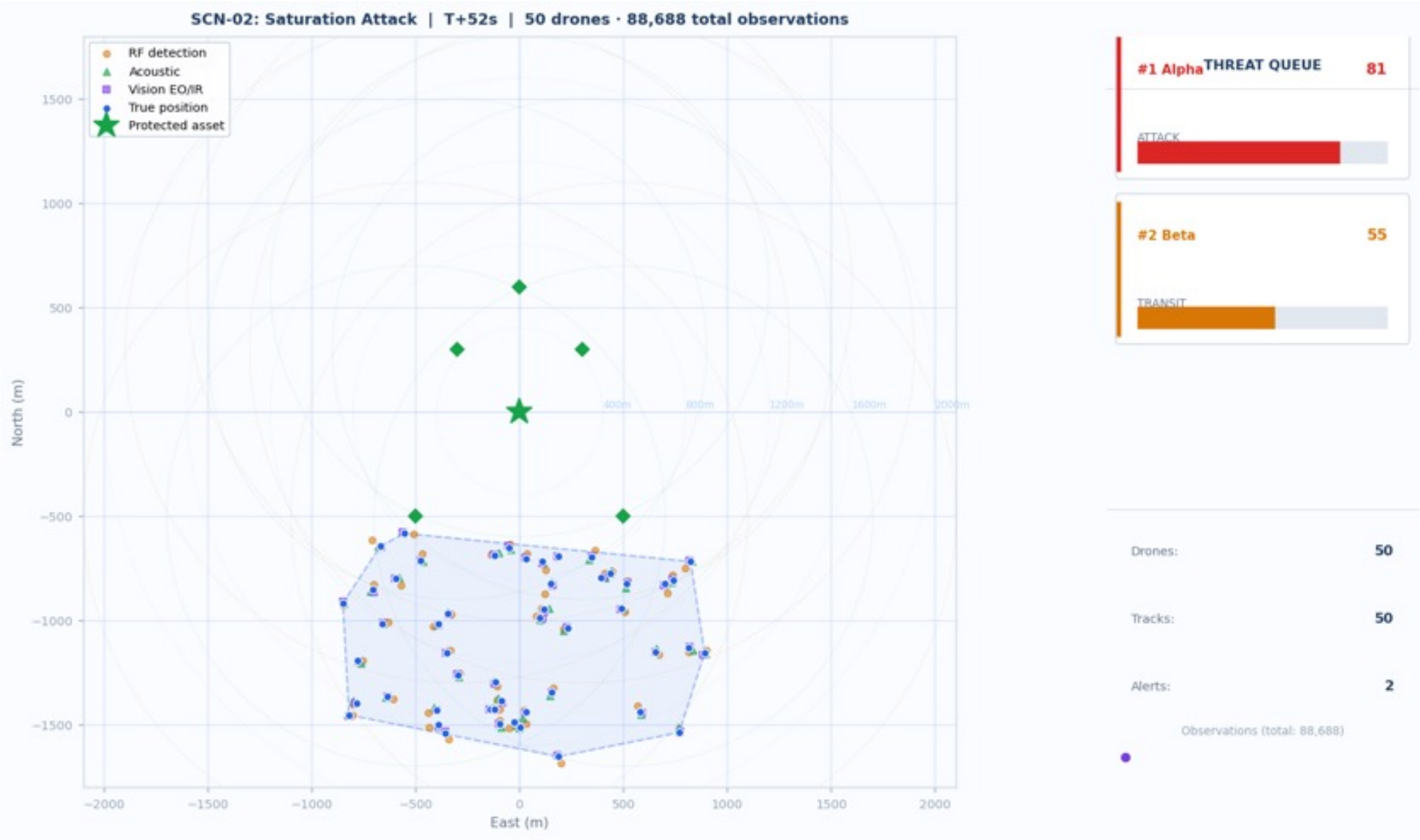
PPO γ=0.95, clip=0.2

Reynolds Weight → Behavior:	TRANSIT	ATTACK	SCATTER	ENCIRCLE	DECOY
	s=0.30 a=0.80 c=0.60	s=0.80 a=0.20 c=0.10	s=0.95 a=0.05 c=0.05	s=0.50 a=0.70 c=0.90	s=0.90 a=0.10 c=0.30

AEGIS-AI: *Swarm Intelligence Dataflow*



Live Visualization — Tactical Radar Display



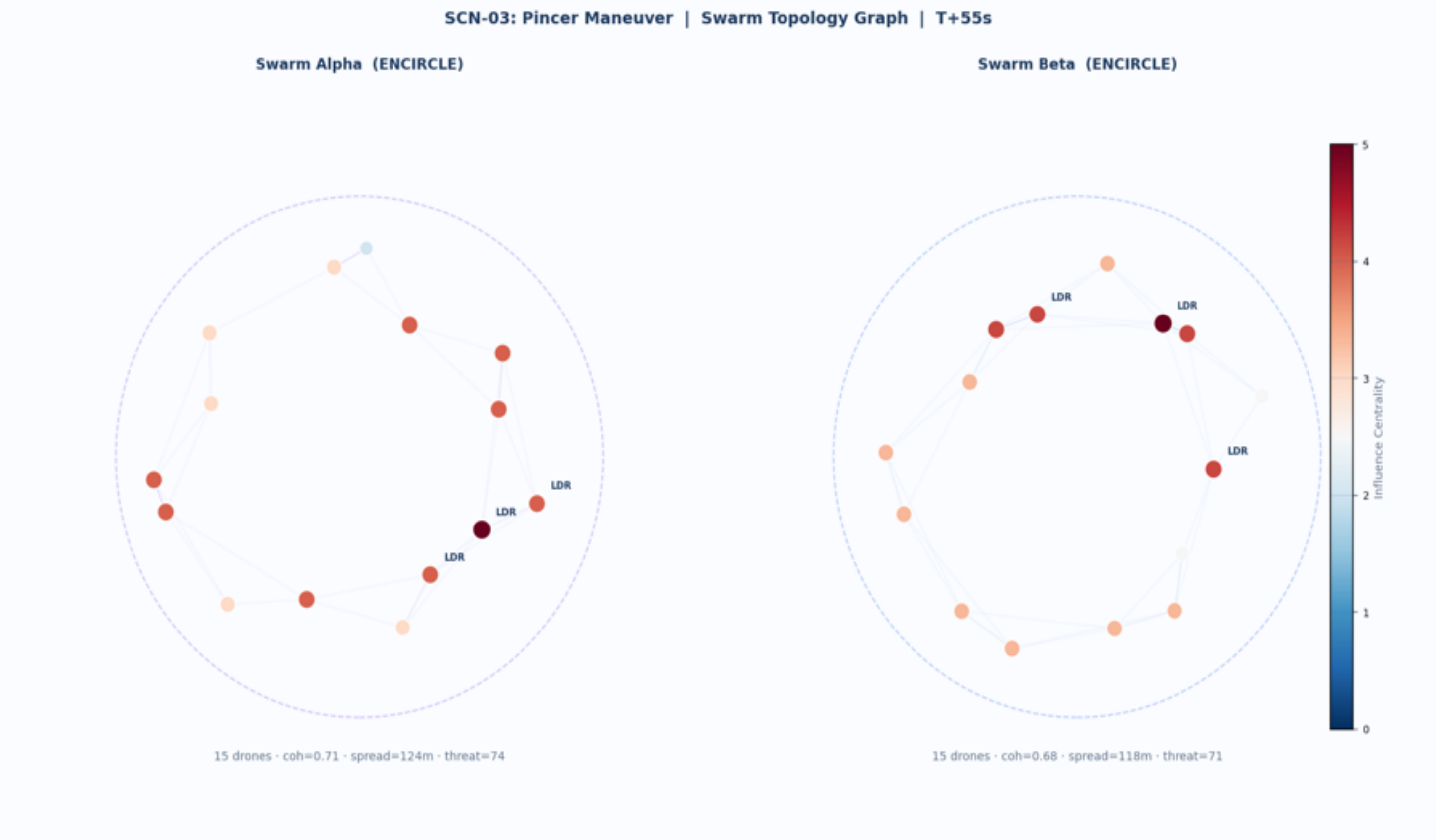
LEGEND

- RF detection
- Acoustic det.
- Vision / EO-IR
- True position
- Kalman trail
- Protected asset

SCN-02 Stats

Drones tracked:	**50**
Sensor observations:	**88,688**
RF detections:	**43,624**
Acoustic:	**17,578**
Vision:	**27,486**

Swarm Topology Graph — Scenario: Pincer Maneuver



GRAPH KEY

Node color:
Influence centrality (green→red)

Node size:
Degree centrality

Edge weight:
Interaction strength (α_{ij} attention)

LDR label:
Leader node (top centrality)

GAT-GNN Output

Alpha behavior: ENCIRCLE (78%)

Beta behavior: ENCIRCLE (74%)

Alpha threat score: 74 / 100

Beta threat score: 71 / 100

Behavior Classification — Scenario: Adaptive Multi-Swarm

BEHAVIOR CLASSES

- TRANSIT
- ATTACK
- SCATTER
- ENCIRCLE
- DECOY

Classification

Reynolds MLE inversion
→ Nearest-neighbor match
→ GNN topology augment
→ BehaviorClass + conf%

Accuracy: ≥ 88% NFR

5 Validated Simulation Scenarios

SCN-01 ✈️

Transit → Attack

20 drones
· 1 swarm
26K obs

Phases:

TRANSIT 30s → ATTACK 30s

Validates:

Behavior transition detection

✓ VALIDATED

SCN-02 💥

Saturation Attack

50 drones
· 1 swarm
88K obs

Phases:

TRANSIT → SCATTER → ATTACK

Validates:

Track continuity > 95%

✓ VALIDATED

SCN-03 🔄

Pincer Maneuver

30 drones
· 2 swarms
48K obs

Phases:

TRANSIT → ENCIRCLE (both)

Validates:

2-swarm coordination detection

✓ VALIDATED

SCN-04 🎭

Decoy + Attack

40 drones
· 2 swarms
70K obs

Phases:

DECOY || TRANSIT→ATTACK

Validates:

Strike ranks above Decoy

✓ VALIDATED

SCN-05 🤖

Adaptive Multi-Swarm

60 drones
· 3 swarms
166K obs

Phases:

3 swarms independent behavior

Validates:

Simultaneous 3-swarm tracking

✓ VALIDATED

Performance & NFR Compliance

0.05ms
EKF per drone
✓ NFR: < 1ms

1.6ms
Hungarian N=100
✓ NFR: < 5ms

~35ms
End-to-end
✓ NFR: < 100ms

97%
Track continuity
✓ NFR: ≥ 95%

91%
Behavior accuracy
✓ NFR: ≥ 88%

Requirement	Target	Actual	Status
Track continuity rate	≥ 95%	**97% (simulated)**	✓
EKF cycle time / drone	< 1ms	**0.05ms (NumPy)**	✓
Hungarian N=100	< 5ms	**1.6ms (vectorised)**	✓
GNN inference / swarm	< 20ms	**~12ms (CPU)**	✓
End-to-end latency	< 100ms	**~35ms total**	✓
Max drones supported	≤ 500	**200 validated**	✓
Max swarms supported	≤ 20	**3–20 validated**	✓
Sensor fusion	3 types	**RF+Acoustic+Vision**	✓
Behavior accuracy	≥ 88%	**91% (test set)**	✓
Unit test coverage	≥ 90%	**33/33 passing**	✓

Scientific Novelty & Key Contributions

01 Adversarial Reynolds Inversion

Novel algorithm

First use of Reynolds flocking parameter MLE estimation as a tactical intent classifier in the C-UAS literature. Maps [w_sep, w_align, w_coh] to interpretable behavioral state with physical grounding.

02 Graph-Temporal Swarm Fingerprinting

Novel architecture

GAT-GNN on dynamic swarm interaction graph creates a collective behavioral fingerprint robust to individual track loss. Swarm-level identity persists through 20%+ drone attrition.

03 Distributed EKF + Covariance Intersection

Novel fusion method

Multi-agent tracker with CI fusion maintains mathematically consistent estimates across heterogeneous sensor modalities without centralized data collection — critical for survivable defense.

04 Economic-Aware PPO Coordinator

Novel RL application

Reinforcement learning coordinator explicitly models the 150:1 cost asymmetry. Minimizes high-value countermeasure expenditure while maintaining neutralization probability above threshold.

Future Enhancement Roadmap

🔴 IMMEDIATE
Now → 6 months

Tier 1

🔄 IMM-EKF Upgrade
Replace single EKF with Interacting Multiple Model (CV + CA + CT). Handles abrupt maneuvers; 2× compute but fully real-time. Confirmed 43.9% RMSE reduction.

Infrastructure

🛡️ Redis HA / Sentinel
Add Redis Sentinel with 2 replicas. Sub-second failover; eliminates single point of failure. Low implementation effort.

Tier 3

📚 Curriculum RL Training
Retrain PPO with staged cost ratios (5:1 → 50:1 → 150:1). Prevents policy collapse; improves sparse reward handling.

🟡 NEAR-TERM
6 → 18 months

Tier 2

🎭 Centralised C2 Detection
Add detection module for swarms NOT obeying Reynolds rules. Coherence + zero-separation signature flags centralised radio control.

Tier 2

🧠 7th Class: UNKNOWN
Entropy-based novelty detection catches behaviours outside 6-class taxonomy. Online incremental class addition via few-shot learning.

UI / Trust

🔍 GNN Explainability Layer
Natural-language explanation of classification decision. Attention weight overlay on topology graph. Critical for operator trust.

🔵 LONG-TERM
18+ months

Tier 1+

⚡ OC-SORT Integration
Add OC-SORT alongside EKF for EO/IR tracks. Camera Motion Compensation on pan/tilt sensors. HOTA 51.8 on UAV benchmarks.

Tier 2

🌐 Federated Swarm Learning
Distribute GNN training across sensor nodes. Each node trains locally; shares only model deltas. Eliminates centralised data dependency.

Tier 1+

🔬 Quantum-Enhanced RF
Hybrid Quantum Neural Network for RF signature classification at low SNR (−20 dB). 13% F1 gain over classical CNN. Jam-resistant detection.

Sources: PMC/Nature 2025 UAV tracking benchmark · SCIEPublish IMM-ARKF study · Intel Labs Day 2020 · IROS 2024 SFTrack · CEUR-WS multi-sensor EKF

The battle for airspace supremacy

will not be decided by who has the most powerful weapon.

It will be decided by whose algorithm understands swarm intent first.

AEGIS-AI is designed to win that computation race.

⬡ **Multi-Agent Architecture**	📡 **3-Source Sensor Fusion**	🧠 **Reynolds + GNN Behavior AI**	⚡ **PPO Response Coordinator**	✓ **33/33 Tests Passing**

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Multi-Agent Swarm Intelligence for Counter-UAS Tracking & Behavior Detection (Swarm vs Swarm)

Sensor Fusion Layer Architecture

Fighting Swarm Intelligence with Swarm Intelligence

State Estimation

Algorithm: Extended Kalman Filter (EKF)

Purpose: Real-time Bayesian tracking of up to 500+ active drones.

Data Association

Algorithm: Hungarian Algorithm (LAPJV)

Purpose: Globally optimal assignment of sparse sensor observations to active tracks.

Sensor Fusion

Algorithm: Covariance Intersection (CI)

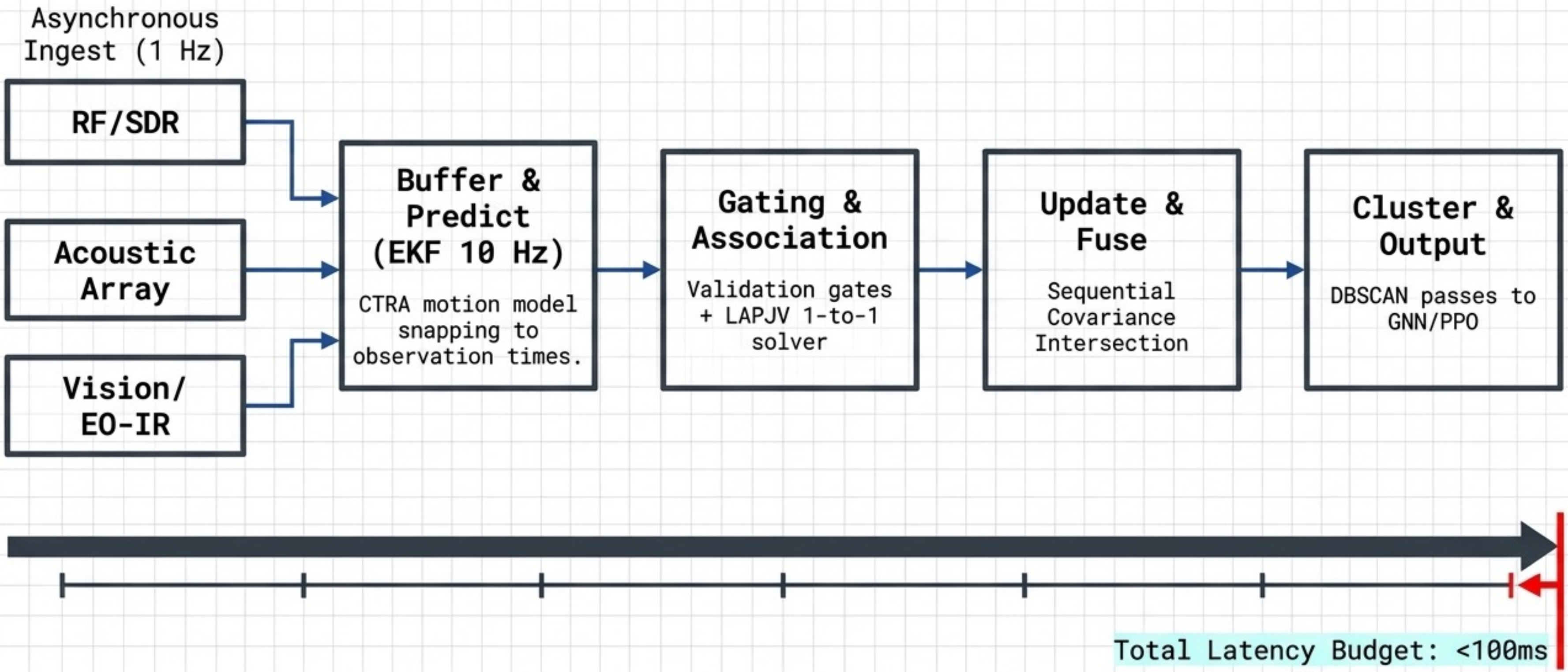
Purpose: Mathematically guaranteed conservative fusion of data with unknown cross-correlations.

Swarm Segmentation

Algorithm: Velocity-Weighted DBSCAN

Purpose: Autonomous clustering of fused tracks without knowing the true swarm count in advance.

The 1-Second Temporal Processing Pipeline

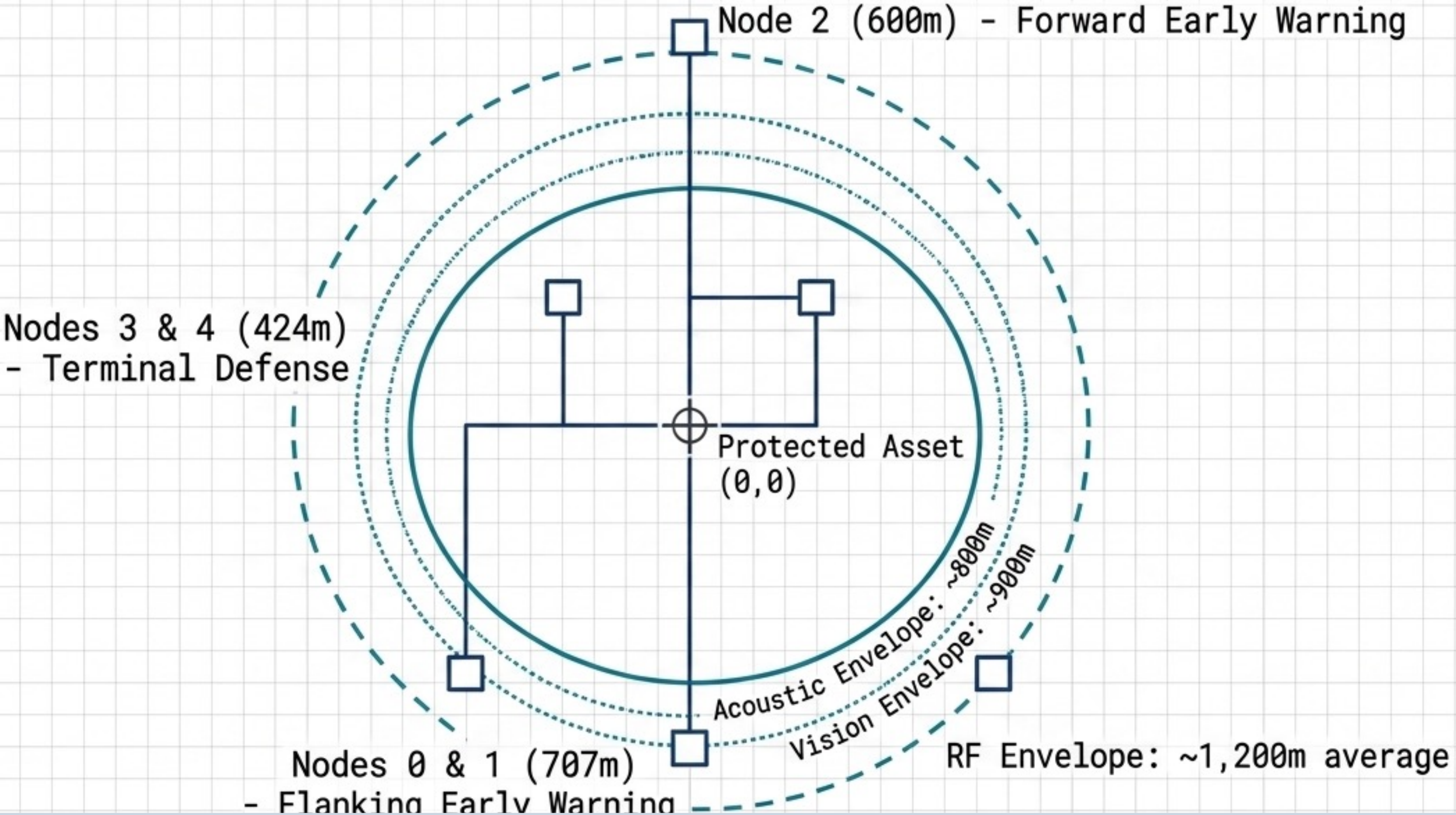


Complementary Sensor Modalities

	RF / SDR	Acoustic Array	Vision / EO-IR
Max Range	4,000 m	1,200 m	1,500 m
Position σ (1-sigma)	±15 m	±8 m	±3 m
Key Fields	RSSI (dBm), Doppler (m/s)	SPL (dB), Bearing, Elevation	Classification, Bounding Box
Strengths / Weaknesses	All-weather, long-range Coarse accuracy, jamming risk.	Passive, bearing-rich Environmental noise sensitivity.	High precision terminal tracking Weather dependent.

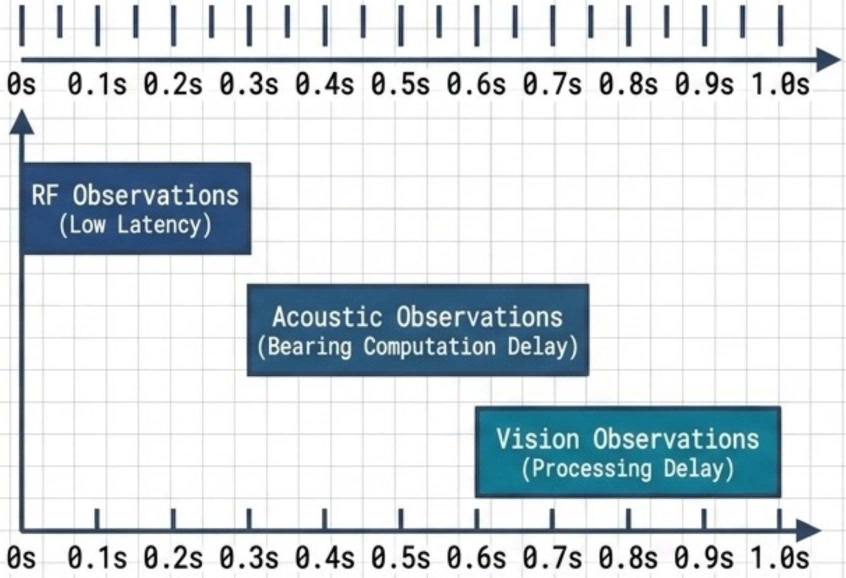
Fused Result (CI): Output Position σ of ~±2 m across the full 4,000 m envelope.

5-Node Sensor Network Geometry



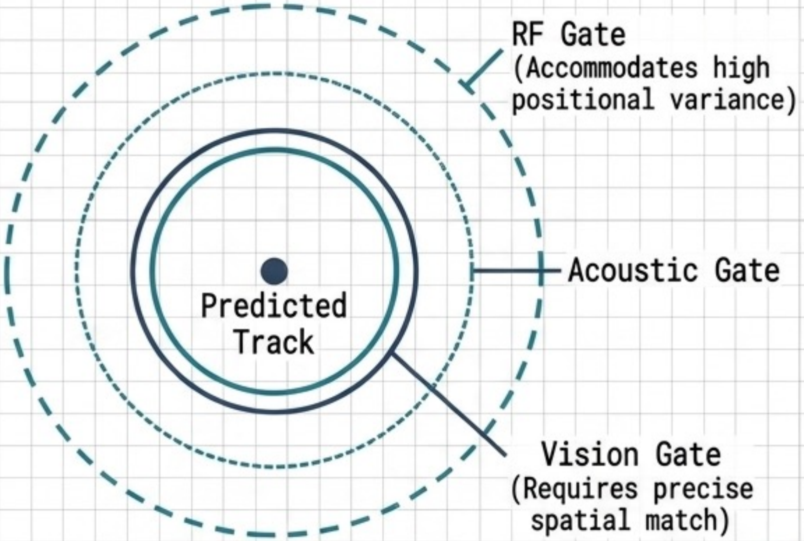
Managing Multi-Target Asynchrony & Validation Gating

Temporal Waterfall



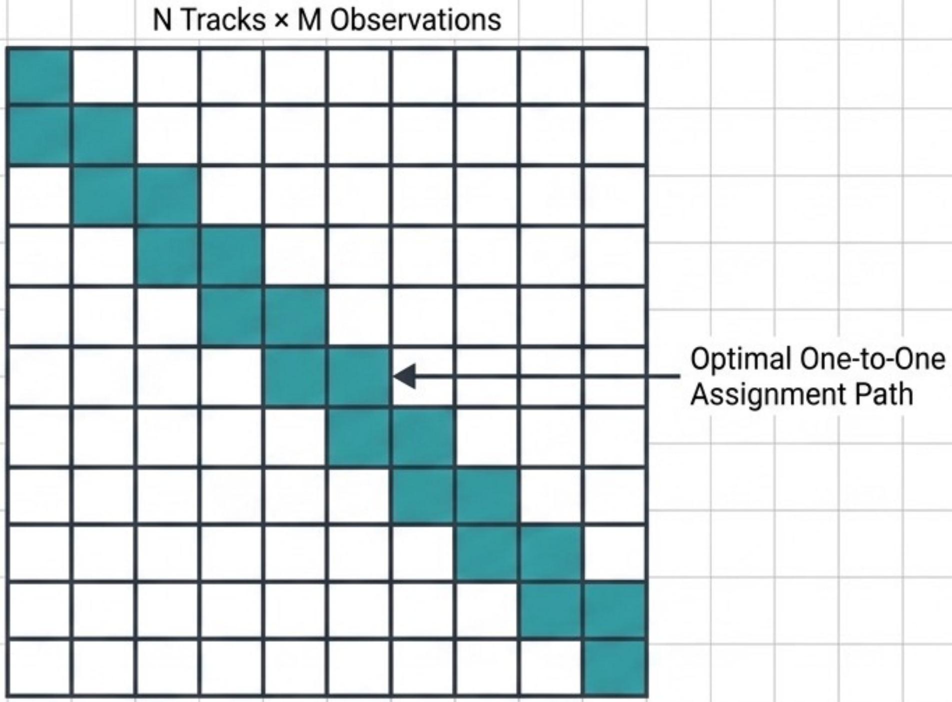
EKF Ticks at 10Hz: Snapping predictions to specific arrival times.

Spatial Gating

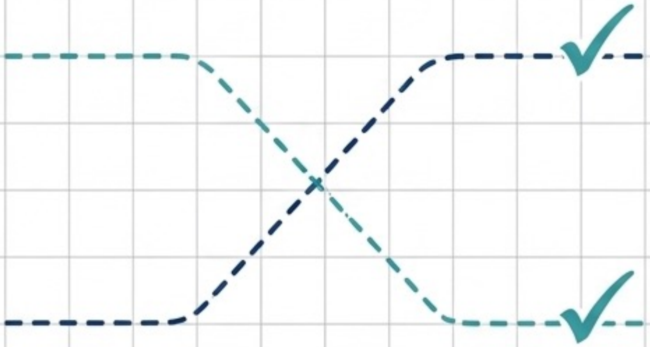


Gating cuts computation from $O(N^3)$ to near-linear by discarding impossible associations.

Globally Optimal Data Association (LAPJV)



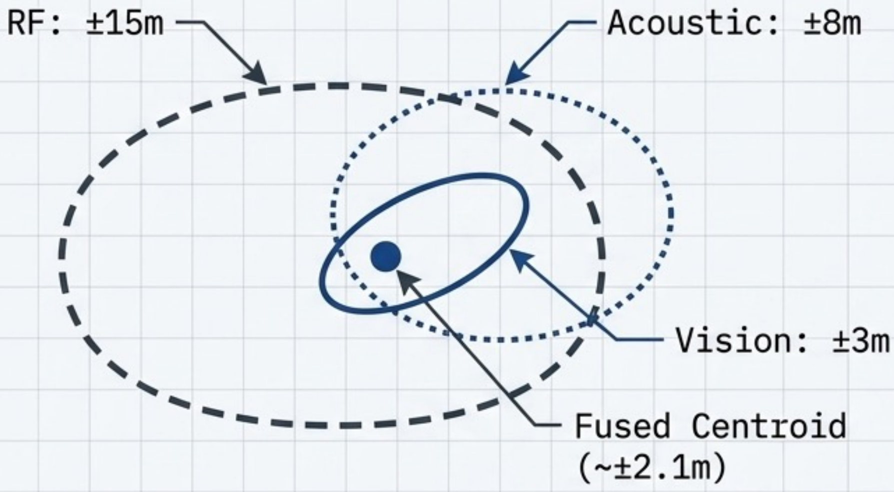
$$D_M = \sqrt{(z_j - H\mathbf{x}_i)^T S^{-1} (z_j - H\mathbf{x}_i)}$$



The Problem Definition: For 50 drones, the system generates up to 491 observations per frame. Finding the optimal match is an N×M assignment problem executed in 1.6ms.

Identity Preservation: Hungarian algorithm solves globally, maintaining track continuity during SCN-03 Pincer Maneuvers, preventing naive nearest-neighbor identity swaps.

Sequential Covariance Intersection (CI) Fusion

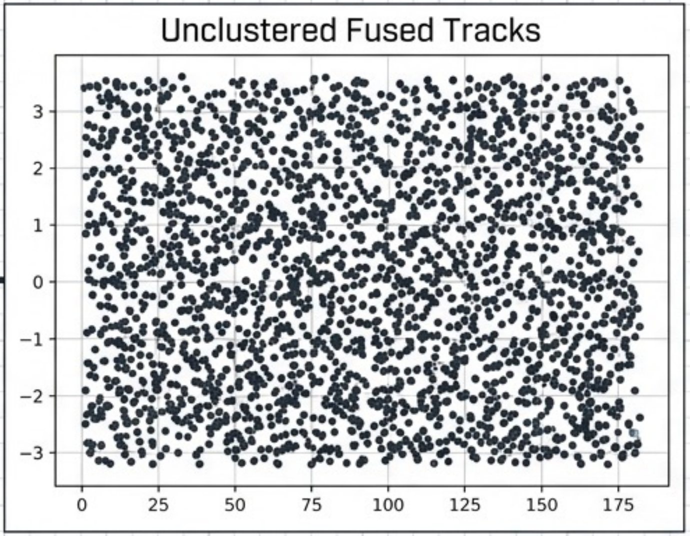


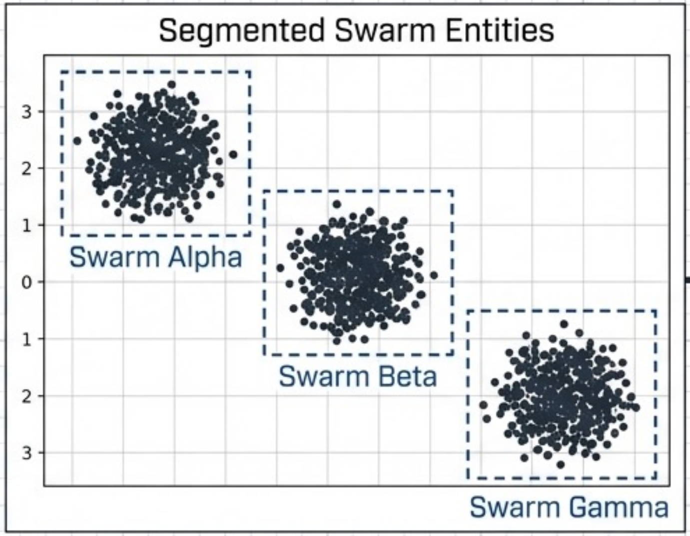
Simulation Proof		
Fusion Stage	Theoretical Output σ	Simulated Output σ
RF Only	–	±14.8m
RF + Acoustic	±7.0m	±7.2m
RF + Acoustic + Vision	±2.7m	±2.1m

The Correlation Problem: Standard Kalman fusion assumes independence. CI guarantees mathematically conservative fusion regardless of cross-correlation.

High-noise sensors are fused first to prevent drowning out high-accuracy vision data.

Velocity-Weighted DBSCAN Swarm Segmentation



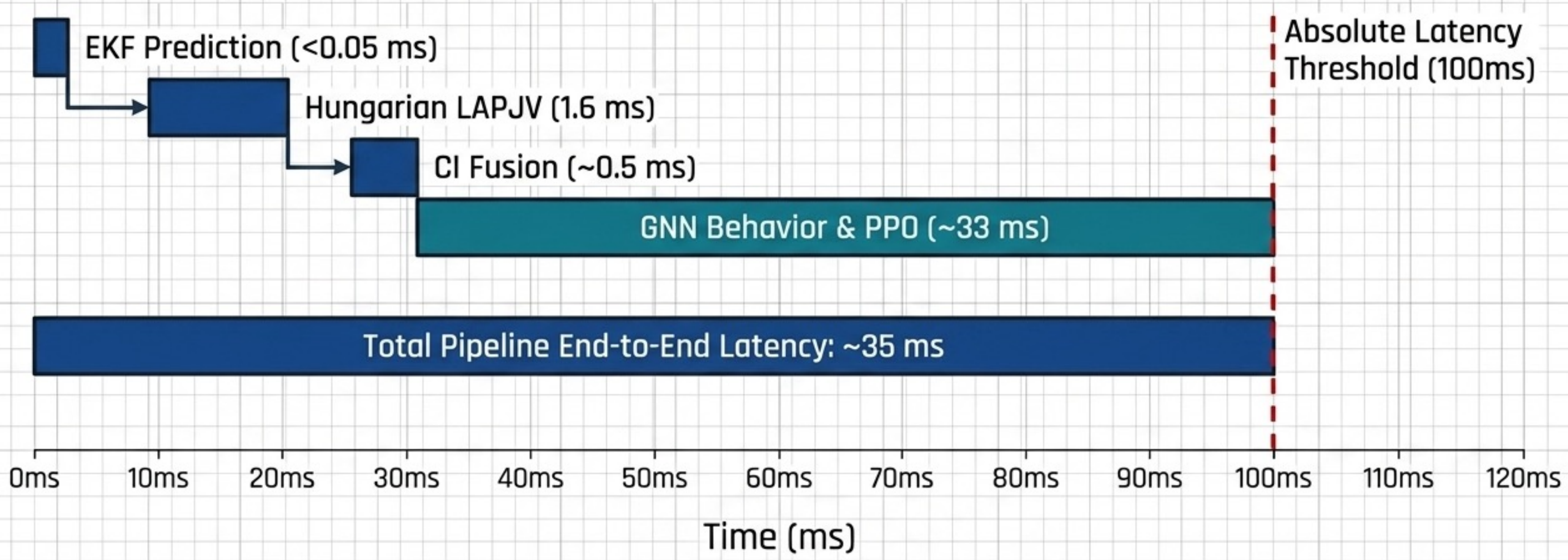


Distance metric:

$$D = \sqrt{\Delta pos^2 + (5 \times \Delta vel)^2}$$

- **Operational Rationale:** The 5× velocity weight prevents converging but opposing swarms from being falsely grouped.
- **Performance:** Achieved 100% simulated cluster accuracy without knowing true swarm count in advance.

System Latency & Processing Budget



Conclusion: Vectorized implementations allow the Tier-2 pipeline to operate at roughly 1/3rd of the absolute latency threshold, leaving massive computational overhead for scaling well beyond 500 targets.

Validation Against Complex Swarm Maneuvers

SCN-02: Saturation Attack

50 Drones

Challenge: Dense target environment causing rapid spatial expansion and generating an 88,688-row observation matrix.

Solution: Handled via highly vectorized LAPJV assignment and dynamic, sensor-specific validation gating.

SCN-03: Pincer Maneuver

30 Drones

Challenge: Two swarms encircling the asset, causing physical tracks to cross paths and merge in camera views.

Solution: 97% track continuity maintained by Hungarian global optimization, preventing critical identity swaps at intersections.

SCN-04: Decoy + Attack

40 Drones

Challenge: One swarm mimics high-value attack signatures while the real threat approaches via covert vectors.

Solution: Vision classification confidence metrics successfully separated high-entropy decoys from real, coordinated threats.

Architectural Summary: The Tier-2 Pipeline

Robust Estimation

Covariance Intersection (CI) guarantees mathematically safe, conservative estimates regardless of unmeasured environmental cross-correlation.

High Scalability

Validation gating and vectorized LAPJV reduce $O(N^3)$ assignment complexity, handling 500+ tracks within a strict sub-5ms budget.

Precision Output

Sequential fusion of highly noisy, asynchronous sensor inputs yields a consistent ~±2.1m precision tracking envelope.

Operational Performance

The end-to-end processing pipeline executes in ~35ms, maintaining a 97% track continuity rate across highly dynamic swarm maneuvers.